

RYAN PRATT

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EDUCATION

University of Wisconsin–Madison

Expected May 2028

B.S. in Computer Engineering; Additional Major in Computer Sciences | GPA: 3.73/4.0

Relevant Coursework: Data Structures & Algorithms, Machine Organization & Programming, User Interfaces

In Progress: Operating Systems, High Performance Computing

SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, C++, SQL, Bash

Frameworks/Libraries: React, Node.js, Express, FastAPI, JUnit, Mockito

Tools/Platforms: Git, Linux, Docker, Maven, SonarQube, Claude Code

Concepts: Data Structures & Algorithms, REST APIs, Unit Testing, Object-Oriented Programming, Full-Stack Development

EXPERIENCE

Wintrust Financial Corporation – Full Stack Engineer Intern

May 2026 – August 2026

- Increased backend test coverage from 16% to 41% across a roughly 76,000-line Java codebase, writing JUnit, Mockito, and AEM Mocks tests for components, services, and servlets.
- Resolved 400+ SonarQube findings, including critical and major issues, by updating production Java code and deprecated APIs, incorporating peer review feedback, and documenting changes for QA.
- Reduced image payload sizes by approximately 30% by developing a Java OSGi service and integrating WebP delivery into custom Adobe Experience Manager components.

iCode – Lead Instructor

June 2025 – August 2025

- Taught 200+ K-12 students programming and robotics through hands-on projects, explaining technical concepts and guiding debugging.

TECHNICAL EXPERIENCE

Wisconsin Robotics – Software Team Member

October 2024 – December 2025

- Implemented A* path planning in Python for rover navigation, reducing time-to-goal by approximately 15% in simulation and physical testing.
- Integrated navigation software for rover deployment, balancing pathfinding performance with hardware resource constraints.

Wisconsin AI Safety Initiative – Technical Safety Scholar

February 2025 – February 2026

- Analyzed approximately 60 AI research papers and model releases during a 13-week cohort, evaluating model capabilities, robustness, and failure modes.
- Compared alignment and interpretability approaches across academic and open-source systems, communicating technical tradeoffs and potential mitigation strategies in peer discussions.

PROJECTS

Azterra – Full-Stack Interactive Atlas

September 2025 – December 2025

Tech: React, Node.js, Express, Leaflet, Supabase, Netlify

- Built a full-stack interactive atlas using React and a Node.js/Express REST API, supporting layered maps, custom markers, and filtering across 34K+ custom map tiles.
- Implemented map editing and filtering, multi-provider authentication, role-based access control, and persistence for 25+ datasets.
- Deployed the application using Netlify serverless infrastructure and developed content ingestion, offline fallback, and backup workflows.

Photon-Sim – Photonic AI Accelerator Simulator

September 2025 – March 2026

Tech: Python, NumPy, SciPy, FastAPI, React, TypeScript

- Built a full-stack simulator with a Python computation core, FastAPI backend, and React/TypeScript dashboard to compare simulated photonic accelerator workloads against CPU baselines.
- Developed a CLI runner to automate benchmarks across matrix sizes 64–1024, exporting reproducible CSV/JSON results for accuracy, latency, and efficiency comparisons.

Amino Acid Substitution Prediction – Machine Learning Marathon 2025

September 2025 – December 2025

Tech: Python, PyTorch, XGBoost, pandas, NumPy, Jupyter Notebook

- Led a five-person team analyzing 175K+ samples across 200 scoresets, guiding preprocessing and feature selection; presented the modeling pipeline and validation strategy to 80+ attendees.
- Diagnosed heavy-tailed prediction errors and redesigned training to prioritize high-error cases, reducing cross-validation mean squared error by 23% and improving internal ranking from third to first.